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Cluster Ranking with an Application to Mining Mailbox Networks

Ziv Bar-Yossef[*,§] Ido Guy[†,‡] Ronny Lempel[‡] Yoëlle S. Maarek[§,¶]
Vladimir Soroka[‡]

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Abstract

We initiate the study of a new clustering framework, called *cluster ranking*. Rather than simply partitioning a network into clusters, a cluster ranking algorithm also orders the clusters by their *strength*. To this end, we introduce a novel strength measure for clusters—the *integrated cohesion*—which is applicable to arbitrary weighted networks.

We then present C-Rank: a new cluster ranking algorithm. Given a network with arbitrary pairwise similarity weights, C-Rank creates a list of overlapping clusters and ranks them by their integrated cohesion. We provide extensive theoretical and empirical analysis of C-Rank and show that it is likely to have high precision and recall.

A main component of C-Rank is a heuristic algorithm for finding sparse vertex separators. At the core of this algorithm is a new connection between the well known measure of vertex betweenness and multicommodity flow.

Our experiments focus on mining *mailbox networks*. A mailbox network is an egocentric social network, consisting of contacts with whom an individual exchanges email. Ties among contacts are represented by the frequency of their co–occurrence on message headers. C-Rank is well suited to mine such networks, since they are abundant with overlapping communities of highly variable strengths. We demonstrate the effectiveness of C-Rank on the Enron data set, consisting of 130 mailbox networks.

1 Introduction

Cluster ranking Clustering and community identification in networks is a vast area of study, spanning over multiple disciplines, such as computer science, sociology, physics, and biology. While the literature is abundant with diverse clustering methods, most of them fall within the same framework [20]: given a network G, find a partition of G that optimizes a predetermined objective function. The various approaches differ in the types of partitions considered (e.g., flat clustering vs. hierarchical clustering), in the assumptions made on the network (e.g., weighted vs. unweighted networks), in the choice of the objective function (e.g., k-means, normalized cuts), and in the techniques used to solve the optimization problem (e.g., agglomerative vs. divisive).

[*]Department of Electrical Engineering, Technion, Haifa 32000, Israel. Email: zivby@ee.technion.ac.il. Supported by the European Commission Marie Curie International Re-integration Grant.
[†]Department of Computer Science, Technion, Haifa 32000, Israel. Email: ido@cs.technion.ac.il.
[‡]IBM Research Lab in Haifa, Haifa 31905, Israel. Emails: {ido,rlempel,vladi}@il.ibm.com.
[§]Google, Haifa Engineering Center, Israel. Emails: {zivby,yoelle}@google.com.
[¶]Done in part when the author was at the IBM Research Lab in Haifa.